

Interval Belief Propagation for Credal Networks with Multi-Valued Variables

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Abstract

Interval Belief Propagation (IBP) is an iterative message-passing algorithm for approximate marginal inference in credal networks. Unlike the 2U and L2U algorithms which are restricted to binary variables, IBP operates natively on multi-valued variables by propagating interval-vector messages. At each iteration, variable nodes tighten their interval bounds by intersecting incoming messages, while factor nodes compute new bounds by optimizing over the extreme points of local credal sets and the corner points of incoming interval messages. The algorithm is inspired by the ApproxLP framework of Antonucci et al. (2015) and the modified belief propagation for Logical Credal Networks (Qian et al., 2021).

1 Preliminaries

1.1 Factor Graph

Given a credal network $\langle G, \mathcal{K} \rangle$, we construct a **factor graph**: a bipartite graph with variable nodes and factor nodes. Each variable X_i in the credal network corresponds to a variable node with cardinality $|D_i|$. Each conditional credal set $K(X_i \mid \text{Pa}(X_i))$ defines a factor node f_i whose scope is $\{X_i\} \cup \text{Pa}(X_i)$. The factor stores the extreme points of $K(X_i \mid \pi)$ for all parent configurations π .

1.2 Interval Messages

Definition 1 (Interval Message). *For a variable X with k states, an **interval message** is a pair of vectors $(\mathbf{l}, \mathbf{u}) \in \mathbb{R}^k \times \mathbb{R}^k$ where $0 \leq l_x \leq u_x \leq 1$ for each state $x \in D_X$. The interval $[l_x, u_x]$ bounds the probability $P(X = x)$ from the message sender’s perspective.*

Messages are of two types:

- $\mu_{v \rightarrow f} = (\mathbf{l}_{v \rightarrow f}, \mathbf{u}_{v \rightarrow f})$: message from variable v to factor f .
- $\mu_{f \rightarrow v} = (\mathbf{l}_{f \rightarrow v}, \mathbf{u}_{f \rightarrow v})$: message from factor f to variable v .

All messages are initialized to the vacuous interval $([0, \dots, 0], [1, \dots, 1])$.

2 Algorithm

Algorithm 1 Interval Belief Propagation (IBP)

Require: Factor graph with extreme points, query Z , evidence $\mathbf{Y} = \mathbf{y}$, max iterations T , threshold τ

Ensure: Bounds $[\underline{P}(Z=z), \overline{P}(Z=z)]$ for each z

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1: Initialize  $\mu_{v \rightarrow f} \leftarrow (\mathbf{0}, \mathbf{1})$  and  $\mu_{f \rightarrow v} \leftarrow (\mathbf{0}, \mathbf{1})$  for all edges
2: for each evidence variable  $Y_j = y_j$  do
3:   Set  $\mu_{Y_j \rightarrow f} \leftarrow (\delta_{y_j}, \delta_{y_j})$  for all factors  $f \in \mathcal{N}(Y_j)$ 
4: end for

5: for  $t = 1, \dots, T$  do
6:    $\delta \leftarrow 0$ 
                                      $\triangleright$  VARIABLE-TO-FACTOR UPDATES
7:   for each variable  $v$  (not evidence), each factor  $f \in \mathcal{N}(v)$  do
8:     for each state  $x$  do
9:        $l_{v \rightarrow f}(x) \leftarrow \max_{f' \in \mathcal{N}(v) \setminus \{f\}} l_{f' \rightarrow v}(x)$ 
10:       $u_{v \rightarrow f}(x) \leftarrow \min_{f' \in \mathcal{N}(v) \setminus \{f\}} u_{f' \rightarrow v}(x)$ 
11:     end for
12:      $\mathbf{u}_{v \rightarrow f} \leftarrow \max(\mathbf{l}_{v \rightarrow f}, \mathbf{u}_{v \rightarrow f})$   $\triangleright$  Ensure  $l \leq u$ 
13:     Update  $\delta$ 
14:   end for
                                      $\triangleright$  FACTOR-TO-VARIABLE UPDATES
15:   for each factor  $f_i$ , each target variable  $v \in \text{scope}(f_i)$  do
16:      $\mathbf{l}_{\text{new}} \leftarrow \mathbf{1}, \quad \mathbf{u}_{\text{new}} \leftarrow \mathbf{0}$ 
17:     for each vertex combination  $\mathbf{vc}$  do
18:       for each corner-point combination  $\mathbf{cc}$  do
19:         Compute unnormalized marginal  $\mathbf{m}(v)$  using  $\mathbf{vc}$  and  $\mathbf{cc}$ 
20:          $\mathbf{p} \leftarrow \mathbf{m} / \sum_x m(x)$ 
21:          $\mathbf{l}_{\text{new}} \leftarrow \min(\mathbf{l}_{\text{new}}, \mathbf{p})$ 
22:          $\mathbf{u}_{\text{new}} \leftarrow \max(\mathbf{u}_{\text{new}}, \mathbf{p})$ 
23:       end for
24:     end for
25:      $\mu_{f \rightarrow v} \leftarrow (\mathbf{l}_{\text{new}}, \mathbf{u}_{\text{new}})$ 
26:     Update  $\delta$ 
27:   end for
28:   if  $\delta < \tau$  then break
29:   end if
30: end for
                                      $\triangleright$  EXTRACT MARGINAL BOUNDS
31: for each state  $z$  do
32:    $\underline{P}(z) \leftarrow \max_{f \in \mathcal{N}(Z)} l_{f \rightarrow Z}(z)$ 
33:    $\overline{P}(z) \leftarrow \min_{f \in \mathcal{N}(Z)} u_{f \rightarrow Z}(z)$ 
34: end for
35: return  $[\underline{P}(z), \overline{P}(z)]$  for each  $z$ 

```

3 Factor-to-Variable Message Computation

The factor-to-variable message from factor f_i (associated with node X_i , parents $\text{Pa}(X_i)$) to target variable $v \in \text{scope}(f_i)$ is computed by scanning over all *vertex combinations* and *corner-point combinations*, as defined below.

Let $\mathcal{O} = \text{scope}(f_i) \setminus \{v\}$ be the other variables. For each $w \in \mathcal{O}$, let $(\mathbf{l}_w, \mathbf{u}_w) = \mu_{w \rightarrow f_i}$ be the incoming message.

3.1 Vertex Combinations (vc)

Factor f_i stores a conditional credal set $K(X_i \mid \pi)$ for each parent configuration $\pi \in \Omega_{\text{Pa}(X_i)}$. Each credal set has a finite number of extreme points (vertices):

$$\text{ext } K(X_i \mid \pi) = \{P_1(X_i \mid \pi), P_2(X_i \mid \pi), \dots, P_{n_\pi}(X_i \mid \pi)\}.$$

Definition 2 (Vertex Combination). A **vertex combination** $\mathbf{vc} = (j_1, j_2, \dots, j_m)$ selects one vertex per parent configuration: vertex j_k from configuration π_k . It specifies a complete conditional distribution $P_{\mathbf{vc}}(X_i \mid \pi_k)$ for every parent configuration.

The total number of vertex combinations is $\prod_{k=1}^m n_{\pi_k}$.

Example. If factor f_A has parents $\{B, E\}$ with 4 parent configurations $(B=0, E=0)$, $(B=1, E=0)$, $(B=0, E=1)$, $(B=1, E=1)$, and each has 2 vertices, then there are $2^4 = 16$ vertex combinations. The combination $\mathbf{vc} = (1, 1, 1, 1)$ selects vertex 1 from every parent configuration, while $\mathbf{vc} = (2, 1, 2, 1)$ selects vertex 2 from $(B=0, E=0)$ and $(B=0, E=1)$, and vertex 1 from the other two.

3.2 Corner-Point Combinations (cc)

For each “other” variable $w \in \mathcal{O}$, the incoming interval message provides a lower bound vector $\mathbf{l}_w$ and an upper bound vector $\mathbf{u}_w$. To compute the factor-to-variable message, we must evaluate the marginal at the extreme points of these intervals.

Definition 3 (Corner-Point Combination). A **corner-point combination** $\mathbf{cc} = (c_1, c_2, \dots, c_{|\mathcal{O}|})$ where each $c_j \in \{0, 1\}$ selects either the lower ($c_j = 0$) or upper ($c_j = 1$) bound vector for each other variable w_j :

$$\mathbf{q}_{w_j} = \begin{cases} \mathbf{l}_{w_j} & \text{if } c_j = 0 \\ \mathbf{u}_{w_j} & \text{if } c_j = 1 \end{cases}$$

The total number of corner-point combinations is $2^{|\mathcal{O}|}$.

Example. When computing $f_A \rightarrow B$ (target is B), the other variables are $\mathcal{O} = \{A, E\}$. If the incoming messages are $\mu_{A \rightarrow f_A} = ([0.7, 0.0], [1.0, 0.3])$ and $\mu_{E \rightarrow f_A} = ([0.9, 0.05], [0.95, 0.1])$, then there are $2^2 = 4$ corner-point combinations selecting $\mathbf{q}_A \in \{[0.7, 0.0], [1.0, 0.3]\}$ and $\mathbf{q}_E \in \{[0.9, 0.05], [0.95, 0.1]\}$.

3.3 Marginal Computation

For a given $(\mathbf{vc}, \mathbf{cc})$ pair, the unnormalized marginal over v is:

Case 1: $v = X_i$ (target is the child node):

$$m(v=x) = \sum_{\pi} P_{\mathbf{vc}}(X_i=x \mid \pi) \cdot \prod_{w \in \text{Pa}(X_i) \cap \mathcal{O}} \mathbf{q}_w(\pi_w), \quad (1)$$

where $\mathbf{q}_w$ is the bound vector for w selected by $\mathbf{cc}$, and π_w is the value of w in parent configuration π .

Case 2: $v \in \text{Pa}(X_i)$ (target is a parent):

$$m(v=x) = \sum_{\pi: \pi_v=x} \left(\sum_{x_i} P_{\mathbf{vc}}(X_i=x_i \mid \pi) \cdot \mathbf{q}_{X_i}(x_i) \right) \cdot \prod_{w \in \text{Pa} \setminus \{v\}} \mathbf{q}_w(\pi_w), \quad (2)$$

where $\mathbf{q}_{X_i}$ is the bound vector for X_i if $X_i \in \mathcal{O}$.

The marginal is normalized: $p(x) = m(x) / \sum_{x'} m(x')$.

The factor-to-variable message then takes the componentwise minimum and maximum over all $(\mathbf{vc}, \mathbf{cc})$ pairs:

$$l_{f \rightarrow v}(x) = \min_{\mathbf{vc}, \mathbf{cc}} p(x),$$

$$u_{f \rightarrow v}(x) = \max_{\mathbf{vc}, \mathbf{cc}} p(x).$$

4 Detailed Running Example: Alarm Network

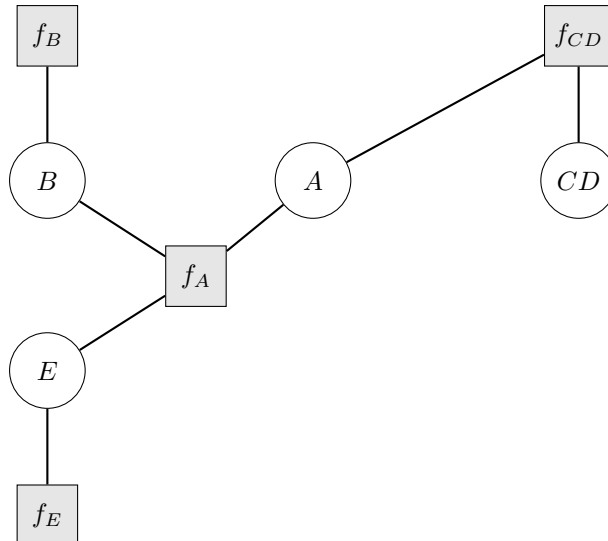
We trace the IBP algorithm on the alarm network from file `alarm.lcn`:

```
s1: 0.1 <= P(B) <= 0.2
s2: 0.05 <= P(E) <= 0.1
s3: 0.6 <= P(A | (B or E)) <= 0.9
s4: 0.5 <= P(not(C xor D) | A) <= 0.7 ; True
s5: 0.05 <= P(A) <= 0.1
```

After the chain graph factorization and LRS vertex enumeration, the credal network has four nodes: B (binary), E (binary), A (binary), and CD (4-valued, representing the compound variable for C and D). The DAG structure is $B \rightarrow A$, $E \rightarrow A$, $A \rightarrow CD$.

4.1 Factor Graph

The factor graph is a bipartite graph with 4 variable nodes and 4 factor nodes:



Each factor's scope and adjacency:

Factor	Node	Parents	Scope
f_B	B	—	$\{B\}$
f_E	E	—	$\{E\}$
f_A	A	B, E	$\{A, B, E\}$
f_{CD}	CD	A	$\{CD, A\}$

Variable adjacency: $B \leftrightarrow \{f_B, f_A\}$, $E \leftrightarrow \{f_E, f_A\}$, $A \leftrightarrow \{f_A, f_{CD}\}$, $CD \leftrightarrow \{f_{CD}\}$.

4.2 Credal Factors (Extreme Points)

Factor f_B : $K(B)$, no parents.

Vertex	$P(B=0)$	$P(B=1)$
v_1	0.80	0.20
v_2	0.90	0.10

Factor f_E : $K(E)$, no parents.

Vertex	$P(E=0)$	$P(E=1)$
v_1	0.90	0.10
v_2	0.95	0.05

Factor f_A : $K(A \mid B, E)$, **parents** $\{B, E\}$. Four parent configurations, each with 2 vertices ($2^4 = 16$ vertex combinations):

Parent config	Vertex	$P(A=0 \mid \cdot)$	$P(A=1 \mid \cdot)$
$B=0, E=0$	v_1	0.9556	0.0444
$B=0, E=0$	v_2	1.0000	0.0000
$B=1, E=0$	v_1	0.0000	1.0000
$B=1, E=0$	v_2	1.0000	0.0000
$B=0, E=1$	v_1	0.0000	1.0000
$B=0, E=1$	v_2	1.0000	0.0000
$B=1, E=1$	v_1	0.0000	1.0000
$B=1, E=1$	v_2	1.0000	0.0000

Factor f_{CD} : $K(CD \mid A)$, **parent** $\{A\}$. Two parent configurations: $A=0$ (4 vertices) and $A=1$ (11 vertices). For brevity, we show a subset:

Config	Vtx	$P(CD=0)$	$P(CD=1)$	$P(CD=2)$	$P(CD=3)$
$A=0$	v_1	0.00	0.00	0.00	1.00
$A=0$	v_2	0.00	0.00	1.00	0.00
$A=0$	v_3	0.00	1.00	0.00	0.00
$A=0$	v_4	1.00	0.00	0.00	0.00
$A=1$	v_1	0.00	0.00	0.30	0.70
$A=1$	v_2	0.00	0.30	0.00	0.70
$\dots$ (9 more)					

There are $4 \times 11 = 44$ vertex combinations for f_{CD} .

4.3 Query: $P(B=1)$, No Evidence

4.3.1 Iteration 1: Factor-to-Variable Messages

Message $f_B \rightarrow B$. Factor f_B has scope $\{B\}$ with no parents and no other variables ($\mathcal{O} = \emptyset$).

- **Vertex combinations:** 2 (one per vertex of $K(B)$, since there is only one parent configuration—the empty one).
- **Corner-point combinations:** $2^0 = 1$ (no other variables).

Computation for each **vc**:

$$\begin{aligned} \mathbf{vc} = (1) : \quad m = v_1 &= [0.80, 0.20], \quad p = m/1.0 = [0.80, 0.20]. \\ \mathbf{vc} = (2) : \quad m = v_2 &= [0.90, 0.10], \quad p = m/1.0 = [0.90, 0.10]. \end{aligned}$$

Result:

$$\mu_{f_B \rightarrow B} = \left(\underbrace{[0.80, 0.10]}_{\mathbf{l}=\min}, \underbrace{[0.90, 0.20]}_{\mathbf{u}=\max} \right).$$

Message $f_E \rightarrow E$. Analogous computation:

$$\mu_{f_E \rightarrow E} = \left([0.90, 0.05], [0.95, 0.10] \right).$$

Message $f_A \rightarrow A$. Factor f_A has scope $\{A, B, E\}$, target is A , so $\mathcal{O} = \{B, E\}$. This is Case 1 (target is the child node). Using Equation (??):

$$m(A=x) = \sum_{\pi \in \{0,1\}^2} P_{\mathbf{vc}}(A=x \mid B=\pi_B, E=\pi_E) \cdot \mathbf{q}_B(\pi_B) \cdot \mathbf{q}_E(\pi_E).$$

In iteration 1, the incoming messages are vacuous: $\mu_{B \rightarrow f_A} = ([0, 0], [1, 1])$, $\mu_{E \rightarrow f_A} = ([0, 0], [1, 1])$.

- **Vertex combinations:** $2^4 = 16$ (2 vertices per each of 4 parent configs).
- **Corner-point combinations:** $2^2 = 4$ (lower or upper for B and E).

When both incoming messages are vacuous, many $(\mathbf{vc}, \mathbf{cc})$ pairs produce $m = [0, 0]$ (zero total) because $\mathbf{q}_B = [0, 0]$ or $\mathbf{q}_E = [0, 0]$ zeroes out the product. Only $\mathbf{cc} = (1, 1)$ (both upper bounds $= [1, 1]$) produces nonzero marginals.

Detailed example: $\mathbf{vc} = (1, 1, 1, 1)$ (vertex 1 for all configs), $\mathbf{cc} = (1, 1)$ ($\mathbf{q}_B = [1, 1]$, $\mathbf{q}_E = [1, 1]$):

$$\begin{aligned} m(A=0) &= 0.9556 \cdot 1 \cdot 1 + 0.0000 \cdot 1 \cdot 1 + 0.0000 \cdot 1 \cdot 1 + 0.0000 \cdot 1 \cdot 1 = 0.9556, \\ m(A=1) &= 0.0444 \cdot 1 \cdot 1 + 1.0000 \cdot 1 \cdot 1 + 1.0000 \cdot 1 \cdot 1 + 1.0000 \cdot 1 \cdot 1 = 3.0444, \\ p &= [0.9556, 3.0444]/4.0 = [0.2389, 0.7611]. \end{aligned}$$

After scanning all 16 vertex combinations (with $\mathbf{cc} = (1, 1)$), the resulting message is wide because vacuous inputs provide no information.

Message $f_A \rightarrow B$. Target is B (a parent), $\mathcal{O} = \{A, E\}$. This is Case 2. With vacuous incoming messages, the message to B is similarly wide.

Messages from f_{CD} . Factor f_{CD} has scope $\{CD, A\}$ with 44 vertex combinations and $2^1 = 2$ corner-point combinations per target variable. In the first iteration with vacuous incoming messages, these produce wide bounds.

4.3.2 Iteration 1: Variable-to-Factor Messages

After the factor-to-variable messages are computed, the variable-to-factor messages are updated by *tightening*: each variable sends to a factor the intersection of messages from all *other* factors.

Message $B \rightarrow f_A$. Variable B has neighbors $\{f_B, f_A\}$. The message to f_A excludes f_A itself:

$$\mu_{B \rightarrow f_A} = \mu_{f_B \rightarrow B} = ([0.80, 0.10], [0.90, 0.20]).$$

This is a significant tightening from the vacuous $([0, 0], [1, 1])$: factor f_B directly constrains B 's marginal, and this information is now passed to f_A .

Message $E \rightarrow f_A$. Similarly, E has neighbors $\{f_E, f_A\}$:

$$\mu_{E \rightarrow f_A} = \mu_{f_E \rightarrow E} = ([0.90, 0.05], [0.95, 0.10]).$$

Message $A \rightarrow f_{CD}$. Variable A has neighbors $\{f_A, f_{CD}\}$:

$$\mu_{A \rightarrow f_{CD}} = \mu_{f_A \rightarrow A}.$$

4.3.3 Iteration 2: Recomputing $f_A \rightarrow A$ with Informative Messages

Now f_A receives informative messages from B and E . The computation of $f_A \rightarrow A$ uses:

$$\begin{aligned} \mathbf{q}_B &\in \{[0.80, 0.10], [0.90, 0.20]\}, \\ \mathbf{q}_E &\in \{[0.90, 0.05], [0.95, 0.10]\}. \end{aligned}$$

Detailed example: $\mathbf{vc} = (1, 1, 1, 1)$, $\mathbf{cc}$: $\mathbf{q}_B = [0.80, 0.10]$, $\mathbf{q}_E = [0.90, 0.05]$:

$$\begin{aligned} m(A=0) &= 0.9556 \cdot 0.80 \cdot 0.90 + 0.0000 \cdot 0.10 \cdot 0.90 \\ &\quad + 0.0000 \cdot 0.80 \cdot 0.05 + 0.0000 \cdot 0.10 \cdot 0.05 = 0.6880, \\ m(A=1) &= 0.0444 \cdot 0.80 \cdot 0.90 + 1.0000 \cdot 0.10 \cdot 0.90 \\ &\quad + 1.0000 \cdot 0.80 \cdot 0.05 + 1.0000 \cdot 0.10 \cdot 0.05 = 0.1670, \\ p &= [0.6880, 0.1670]/0.8550 = [0.8047, 0.1953]. \end{aligned}$$

Another example: $\mathbf{vc} = (1, 1, 1, 1)$, $\mathbf{cc}$: $\mathbf{q}_B = [0.90, 0.20]$, $\mathbf{q}_E = [0.95, 0.10]$:

$$\begin{aligned} m(A=0) &= 0.9556 \cdot 0.90 \cdot 0.95 + 0.0 \cdot 0.20 \cdot 0.95 \\ &\quad + 0.0 \cdot 0.90 \cdot 0.10 + 0.0 \cdot 0.20 \cdot 0.10 = 0.8170, \\ m(A=1) &= 0.0444 \cdot 0.90 \cdot 0.95 + 1.0 \cdot 0.20 \cdot 0.95 \\ &\quad + 1.0 \cdot 0.90 \cdot 0.10 + 1.0 \cdot 0.20 \cdot 0.10 = 0.3380, \\ p &= [0.8170, 0.3380]/1.1550 = [0.7074, 0.2926]. \end{aligned}$$

As the incoming messages become more informative (tighter intervals), the factor explores a more constrained space and the message to A tightens accordingly.

4.3.4 Final Marginal Extraction

After convergence, the marginal bounds for B are:

$$\underline{P}(B=x) = \max_{f \in \{f_B, f_A\}} l_{f \rightarrow B}(x), \quad \overline{P}(B=x) = \min_{f \in \{f_B, f_A\}} u_{f \rightarrow B}(x).$$

Factor f_B provides the tightest bounds for B since it directly encodes $K(B)$:

$$P(B=1) \in [0.10, 0.20].$$

4.4 Comparison with Exact Variable Elimination

Query	Method	$\underline{P}$	$\overline{P}$
$P(B=1)$	Exact VE	0.1000	0.2000
$P(B=1)$	IBP	0.1000	0.2000
$P(A=1 \mid B=0, E=0)$	Exact VE	0.0000	0.0444
$P(A=1 \mid B=0, E=0)$	IBP	0.0000	0.0444

For root nodes like B , IBP recovers the exact bounds because the local factor alone determines the marginal. For conditional queries with evidence, the bounds may differ on loopy graphs due to the approximation inherent in message passing.

5 Multi-Valued Variable Support

A key advantage of IBP over L2U and GL2U is native support for multi-valued variables:

- **L2U**: Restricted to binary variables. Messages are scalar intervals.

- **GL2U**: Handles multi-valued variables by binarization (replacing a k -valued variable with $\lceil \log_2 k \rceil$ binary variables in a chain). This introduces auxiliary structure and can be lossy.
- **IBP**: Messages are interval vectors $(\mathbf{l}, \mathbf{u}) \in \mathbb{R}^k \times \mathbb{R}^k$. No binarization needed. Each state is bounded independently.

For example, the compound variable CD in the alarm network has 4 states. IBP directly propagates 4-dimensional interval vectors for CD , while GL2U would require binarizing CD into two binary variables.

6 Convergence Properties

- Messages are updated by taking min/max over all vertex and corner-point combinations, so the algorithm explores the full range of the credal set.
- On polytree-shaped networks with binary variables, IBP reduces to the 2U algorithm and converges to exact bounds.
- On loopy graphs, convergence to a fixed point is empirically observed but not formally guaranteed. The fixed point may not correspond to the exact bounds (similar to loopy belief propagation for Bayesian networks).

7 Complexity

Each iteration of IBP takes $O(N_f \cdot V_{\max} \cdot 2^{S_{\max}} \cdot k^{S_{\max}})$ time, where N_f is the number of factors, $V_{\max}$ is the maximum number of vertex combinations per factor, $S_{\max}$ is the maximum scope size, and k is the maximum domain size. The number of iterations until convergence depends on the network structure and is typically 10–100 for practical networks.

8 References

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